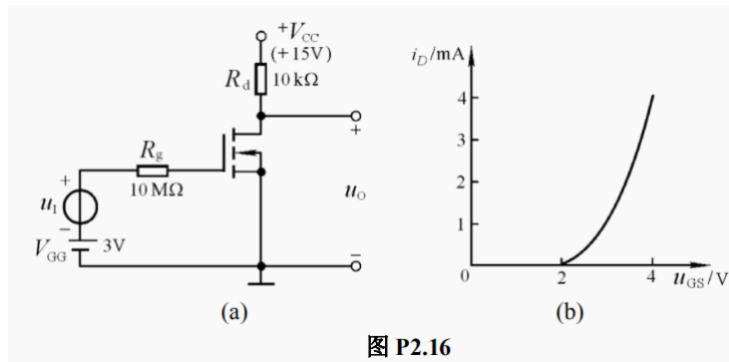


## Homework 4

2.16 已知图 P.16(a)所示电路中场效应管的转移特性如图(b)所示。求解电路的静态工作点 Q 点和交流电压增益  $A_u$ 。

2.16 The transfer characteristics of the FET is shown in figure P2.16 (b). Ask: what's the quiescent working point Q and the voltage gain  $A_u$  of the circuit.



2.17 电路如图 P2.17 所示。(1) 若输出电压波形底部失真，则可采取哪些措施？

若输出电压波形顶部失真，则可采取哪些措施？(2) 若想增大  $|\dot{A}_u|$ ，则可采取哪些措施？

2.17 the circuit is shown in figure p2.17. (1) If the bottom of output voltage waveform is distorted, what measures can be taken? If the top of output voltage waveform is distorted, what measures can be taken? (2) If you want to increase the  $|\dot{A}_u|$ , what measures can be taken?

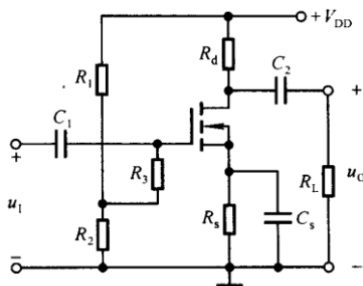
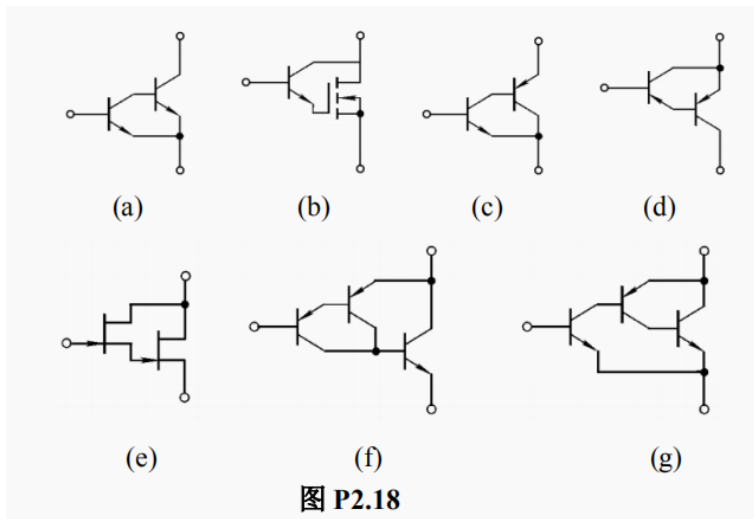


图 P2.17

2.18 图 P2.18 中的哪些接法可以构成复合管?标出它们等效管的类型(如 NPN 型、PNP 型、N 沟道结型.....)及管脚(b、e、c、d、g、s)。

2.18 Which connection in figure P2.18 can make a good composite amplifier? For these good composite amplifier, please show their types (such as NPN type, PNP type, N-type FET, or

P-type FET) and mark the three terminals/pins (b、 e、 c、 d、 g、 s).



3.2 Assume point Q is suitable, draw out AC path, and list  $A_u$ ,  $R_i$ ,  $R_o$ .

**3.2** 设图 P3.2 所示各电路的静态工作点均合适，分别画出它们的交流等效电路，并写出  $\dot{A}_u$ 、 $R_i$  和  $R_o$  的表达式。

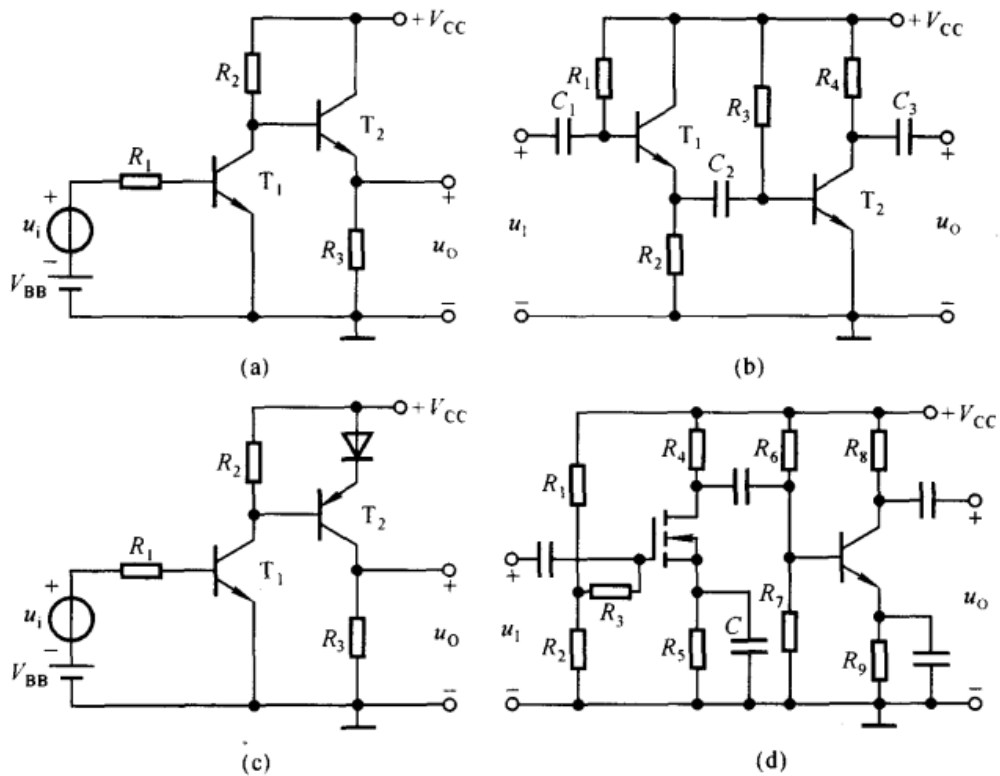
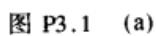


图 P3.2

3.4 As shown in the fig3.1(a),  $\beta=150$ ,  $r_{be}=2k\Omega$ , please calculate  $\dot{A}_u$ ,  $R_i$  and  $R_o$

$\dot{A}_u$ 、 $R_i$ 和 $R_o$ 。 $R_i$  when the probe is shifted to the center of  $R_w$ .

试计算  $R_w$  滑动端在中点时  $T_1$  管和  $T_2$  管的发射极静态电流  $I_{EQ}$ ，以及动态参数  $A_d$  和  $R_i$ 。



3.7 As shown in the fig3.7,  $\beta = 140$ ,  $r_{be} = 4k\Omega$ ,  $u_{i1} = 20mV$ ,  $u_{i2} = 10mV$ , please calculate  $u_{ic} = ?$   $u_{id} = ?$

$$\Delta u_0 = ?$$

3.7 电路如图 P3.7 所示，已知  $T_1$  管和  $T_2$  管的  $\beta$  均为 140， $r_{be}$  均为  $4\text{ k}\Omega$ 。试问：若输入直流信号  $u_{i1} = 20\text{ mV}$ ， $u_{i2} = 10\text{ mV}$ ，则电路的共模输入电压  $u_{ic} = ?$  差模输入电压  $u_{id} = ?$  输出动态电压  $\Delta u_o = ?$

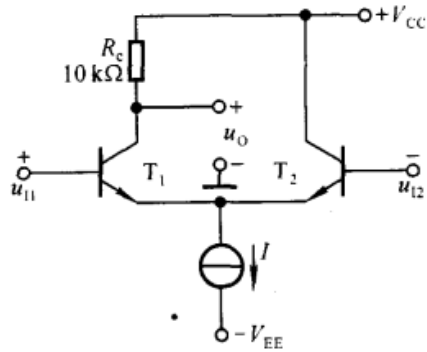


图 P3.7

3.8  $g_m = 10\text{ mS}$ ,  $A_d = ?$ ,  $R_i = ?$

3.8 电路如图 P3.8 所示， $T_1$  和  $T_2$  的低频跨导  $g_m$  均为  $10\text{ mS}$ 。试求解差模放大倍数和输入电阻。

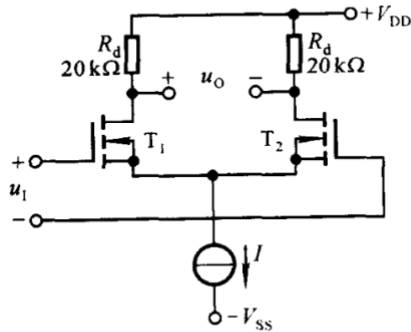


图 P3.8